

Signal Processing–Driven Real-Time Rough Road Detection with GPS Accuracy Improvement on Edge Microcontrollers

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Abstract—The deteriorating condition of road surfaces in developing countries poses significant risks and compromises the safety of participating vehicles. To support authorities in road maintenance, this study proposes an IoT-oriented inertial sensing framework capable of real-time pavement classification and pothole detection, aimed at deployment on low-cost devices. The system integrates an abnormality-index-based algorithm to identify localized damages, combined with a GPS position interpolation procedure to address delayed or missing coordinates. Experimental results demonstrate that this solution significantly improves vehicle position estimation, reducing the mean localization error under 3 m at driving speeds below 50 km/h.

Index Terms—Indoor positioning system, HAR, sensor, microcontroller, signal processing

I. INTRODUCTION

The rapid process of urbanization has led to the deterioration of infrastructure, particularly road systems that degrade over time. At the same time, the rapid development of Intelligent Transportation Systems (ITS) has increased the demand for monitoring road surface quality and detecting road surface defects. Early detection of rough road segments and potholes enables authorities to implement timely maintenance actions and improve traffic safety. However, the deployment of such monitoring systems on a large scale still faces significant challenges, including high equipment costs and limited reliability of positioning data in complex urban environments.

Road networks play an important role in connecting residential, commercial, and industrial areas; therefore, ensuring smooth traffic flow is essential. Traffic conditions are influenced by multiple factors, such as vehicle density, weather conditions, and road surface conditions. Among these factors, the condition of the road surface directly affects traffic safety. Road surfaces should be maintained in good condition to reduce the risk of accidents. Therefore, continuous road maintenance is necessary to mitigate traffic accident risks. Delayed repair of road surface damage

can accelerate structural degradation, increase maintenance costs, and create safety risks for road users.

Existing studies on road surface condition detection generally focus on two main approaches: image-based methods using cameras and inertial sensor–based methods. Image-based approaches can provide detailed information about road surface characteristics; however, they require high computational resources and are sensitive to lighting and weather conditions. In contrast, vibration-based approaches using inertial sensors provide advantages in terms of lower cost and continuous operation under various environmental conditions. A major challenge of this approach lies in handling signal noise generated by vehicle engines, the dependence of vibration signals on vehicle speed, and the accumulated positioning error of the Global Positioning System (GPS) when vehicles move through obstructed areas or dense urban environments.

The objective of this study is to develop a road surface quality warning system based on acceleration data collected from sensors mounted on vehicles. Specifically, the proposed system utilizes data collected from public transportation vehicles or specialized vehicles to automatically detect locations with poor road surface conditions, such as rough segments and potholes, and transmit their positions to the responsible authorities in real time. The study also describes the experimental conditions within the scope of the research, including factors affecting vehicle speed and traffic conditions. An important component of this study is the processing and mitigation of GPS positioning errors in order to improve system accuracy in practical scenarios. In addition, the system is designed with scalability in mind, allowing deployment on a large scale to support road monitoring and maintenance more effectively.

II. RELATED WORKS

The classification of road surface types using inertial sensors and ML is a critical component of ITS, en-

hancing safety and efficiency. This approach is crucial for enhancing road safety and maintenance by providing real-time data on road conditions [1]–[3]. Various ML models, including DL and traditional algorithms, have been employed to classify road surfaces based on data from accelerometers, gyroscopes, and global positioning system (GPS) sensors [4]. Both CML and DL approaches have been explored for this task, each offering unique advantages. CML methods often require manual feature extraction, while DL models can automatically extract features, potentially leading to higher accuracy [5]. The following sections detail the methodologies and findings from recent research in this area.

One early attempt to enhance performance was proposed by Hnoohom *et al.* [6], who introduced the SE-ResNet architecture, a residual CNN augmented with Squeeze-and-Excitation (SE) modules. The reported accuracy reached 98.41 %, although this result was obtained using 5-fold CV on the entire dataset rather than the context-based evaluation protocol. As a result, the comparison with the 93.17 % baseline should be interpreted with caution.

Recurrent neural network (RNN) variants have also been explored to address temporal dependencies in inertial signals. Mekruksavanich *et al.* [7] proposed ResBiGRU-SE, a hybrid architecture combining residual connections, bidirectional gated recurrent units (BiGRU), and SE modules. The reported accuracy was 98.41 %, but similar to previous work, the evaluation relied on 5-fold CV, making it difficult to assess robustness across different vehicles, drivers, and environments.

Aslam *et al.* [8] applied a Transformer-based architecture to the task, using multi-head attention to model long-range temporal dependencies. The method incorporated a Random Forest-based feature selection step to reduce input dimensionality. Using an 80/20 split on PVS 1–8, the model achieved a weighted F1-score of 97 %. When evaluated on an unseen context (PVS 9), performance decreased to 80.41 %, illustrating the sensitivity of such models to contextual variation and highlighting the importance of context-aware evaluation protocols.

Cui *et al.* [9] extended this line of work by proposing a VGG-inspired 1D-CNN architecture achieving 99.3 % accuracy with 101.6k parameters. This result was obtained using a simple 80/20 split. Therefore, while the architecture is compact and efficient, its generalization capability under unseen operational conditions remains unverified relative to the PVS benchmark.

A parallel line of research has focused on improving practicality and computational efficiency. The performance sensitivity observed in several deep learning models, together with the substantial drop reported by Aslam *et al.* [8] when evaluated on an unseen context, further underscores the need for approaches that maintain accuracy while operating reliably across diverse conditions and hardware constraints.

Cong *et al.* [10] revisited tree-based ML models to re-evaluate their potential for RSC. The authors examined Random Forest, Gradient Boosting (GB), and XGBoost, and proposed a Hybrid Filter-Wrapper (HFW) feature selection strategy that combines the efficiency of filter-based ranking with the effectiveness of wrapper-based selection. Their highest reported accuracy was 94.2 %, obtained using an XGBoost model with a substantially reduced feature subset, shrinking the original 126 features to 30 through HFW-GB. This reduction enhanced computational efficiency and training speed, demonstrating that CML methods remain competitive when supported by well-designed feature engineering. However, the 94.2 % result was obtained using a 70/30 split, which differs from the context-aware evaluation protocol defined in [11]. While this limits direct comparability with the 93.17 % baseline, the findings nonetheless indicate that earlier CML performance limitations were influenced by feature design rather than intrinsic model capacity. These results suggest that CML models remain a viable and efficient direction, particularly when computational constraints or embedded deployment requirements are present.

III. MATERIALS AND METHODS

A. System Architecture

As illustrated in Fig. 1, the proposed system is a compact IoT-enabled device that acquires, processes, and transmits road surface information in real time. The sensor node is enclosed in a $7.3 \times 4.5 \times 3.4 \text{ cm}^3$ protective casing and is built around an ESP32 MCU, which executes an embedded CML model on vibration data collected from a GY-85 9-axis IMU. A GPS module (ATGM336H) provides localization, while a SIMCom A7680C modem enables LTE-based data transfer. The unit operates on a 3.7 V–2000 mAh Li-Po battery, allowing self-contained in-vehicle deployment. The system aims to provide road management authorities with timely information on pavement conditions to support maintenance and repair planning. In addition, severely degraded road segments containing potholes or deep depressions are detected and their precise locations are reported in real time, enabling other vehicles to receive immediate hazard warnings.

1) *Private dataset*: The private dataset¹ was collected using a fleet of vehicles dedicated to road inspection tasks. The sensor node was rigidly mounted at the geometric center of each vehicle to capture stable vertical and longitudinal vibration patterns, with a consistent coordinate convention across all recordings (x-axis leftward, y-axis forward, and z-axis downward). The dataset includes tri-axial accelerometer, gyroscope, and magnetometer signals, together with GPS latitude, longitude, timestamp, and speed. Road segments were manually annotated into dirt, cobblestone, and asphalt categories. In total, the recordings

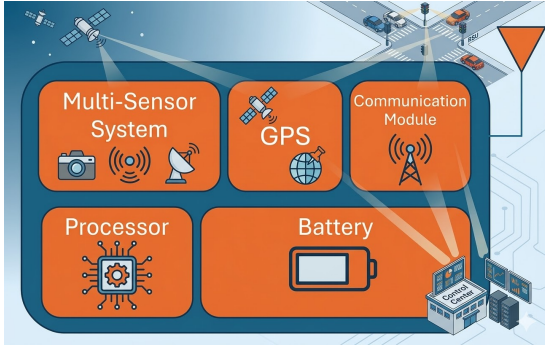


Fig. 1. The proposed system is designed to support specialized traffic service vehicles.

amount to 903.90 minutes (15.06 hours), corresponding to approximately 196 MB of raw measurements.

B. Dataset

A sliding window of 600 samples (6 seconds at 100 Hz) with 50 % overlap was applied to segment the continuous sensor signals into fixed-length windows. This configuration balances temporal coverage and real-time constraints: although each window spans 6 seconds of data to capture richer temporal dynamics, the 300-sample stride ensures that new predictions are produced every 3 seconds, consistent with the real-time latency requirement. Applying this scheme to the private dataset yielded 9039 windows, consisting of 3258 dirt, 2157 cobblestone, and 3624 asphalt segments.

C. Feature engineering

A feature engineering stage was applied to convert each segmented window into a compact representation suitable for training efficiency and embedded deployment. As the design targets real-time operation on resource-constrained hardware, only time-domain features were considered [10]–[12]. For each inertial channel, thirteen statistical descriptors were extracted to characterize central tendency, dispersion, distribution shape, and short-term temporal dynamics, including mean, standard deviation (SD), root mean square (RMS), maximum, minimum, range, median, interquartile range (IQR), skewness, kurtosis, Hjorth mobility, Hjorth complexity, and autocorrelation. Because vibration intensity varies with vehicle speed [11], the mean and SD of the speed signal were also added. This procedure produces a concise feature vector for each 6-second window, summarizing the vibration characteristics while maintaining compatibility with the computational constraints of embedded inference [10], [12].

D. Constrained hyperparameter optimization

Across all experiments, the hyperparameter search space is intentionally restricted to settings suitable for real-time embedded applications. The grid explored includes

`n_estimators` in the range 1–50 and `max_depth` in the range 1–20, which represent the upper bounds compatible with on-device deployment [12]. All four tree-based classifier families (XGBoost, Random Forest–RF, Light Gradient Boosting Machine–LGBM, and Extra Trees–ET) are evaluated under this same constrained search space. For each candidate configuration, a model is trained on the designated training fold and subsequently exported to a C/C++ header file using the `micromlgen` or `m2cgen` library [13]. The resulting header size is then measured to verify compliance with the strict memory budget imposed by the target MCU.

E. GPS-based trajectory interpolation

During data acquisition, GPS receivers occasionally fail to update their coordinates for several consecutive samples due to satellite occlusion, multipath effects, or adverse environmental conditions. These periods produce repeated latitude–longitude pairs despite continuous vehicle motion, creating artificial plateaus in the recorded trajectory. Such artifacts distort travelled-distance measurements, degrade the temporal alignment between IMU and GPS streams, and reduce the reliability of downstream analyses, including anomaly localization.

To mitigate these issues, a velocity-aware interpolation procedure is applied to reconstruct geographically consistent intermediate positions. The method assumes short-term continuous motion and estimates the travelled distance from the instantaneous speed v and elapsed time t between two valid GPS fixes:

$$S = v \times t \quad (1)$$

where S denotes the distance covered during the interval without coordinate updates. A great-circle navigation model is then employed with Earth radius $R = 6371000$ m. Given two valid observations (ϕ_1, λ_1) and (ϕ_2, λ_2) , the bearing angle θ is computed as:

$$\theta = \text{atan2} \left(\frac{\sin(\Delta\lambda) \cos(\phi_2), \cos(\phi_1) \sin(\phi_2)}{-\sin(\phi_1) \cos(\phi_2) \cos(\Delta\lambda)} \right) \quad (2)$$

with $\Delta\lambda = \lambda_2 - \lambda_1$. The interpolated geographic position $(\text{lat}_2, \text{lon}_2)$ at distance S from $(\text{lat}_1, \text{lon}_1)$ along direction θ is then determined as:

$$\begin{cases} \text{lat}_2 = \arcsin \left(\sin(\text{lat}_1) \cos \frac{S}{R} + \cos(\text{lat}_1) \sin \frac{S}{R} \cos \theta \right) \\ \text{lon}_2 = \text{lon}_1 + \arctan 2 \left(\sin \theta \sin \frac{S}{R} \cos(\text{lat}_1), \cos \frac{S}{R} - \sin(\text{lat}_1) \sin(\text{lat}_2) \right) \end{cases} \quad (3)$$

The interpolation algorithm scans the chronologically ordered GPS samples, detects intervals in which latitude and longitude remain unchanged within a small angular tolerance ε , and inserts intermediate points whenever the

¹<https://dx.doi.org/10.21227/x6av-va13>

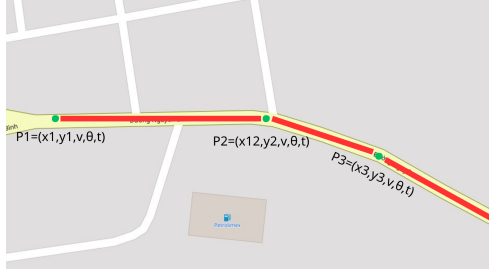


Fig. 2. Illustration of the GPS interpolation algorithm, where P_i are valid GPS points and intermediate points are reconstructed using velocity v and bearing angle θ

instantaneous speed is strictly positive. For each such interval, interpolated positions are generated at a prescribed sampling rate using (1)–(3). The bearing angle is estimated from the nearest available GPS fix, ensuring that transitions between successive road segments follow the correct directional trend. Fig. 2 illustrates the reconstruction principle.

Algorithm 1 summarizes the complete procedure. Only intervals with frozen GPS coordinates are modified; all valid GPS fixes remain unchanged. Because the bearing angle is recalculated whenever new observations become available, the reconstructed trajectory naturally follows curved paths rather than imposing piecewise-linear interpolation.

Algorithm 1: Velocity-aware GPS interpolation

Input: GPS samples (lat, lon, v, t) ; interpolation rate hz .
Output: Interpolated sequence $\mathcal{L}$.

```

1  $step \leftarrow 1/hz$ ;
2  $\mathcal{L} \leftarrow \emptyset$ ;
3  $i \leftarrow 1$ ;
4 while  $i \leq N$  do
5   Find  $j > i$  such that  $GPS(j) \neq GPS(i)$  within tolerance
6   Append sample  $i$  to  $\mathcal{L}$ 
7   if  $j > N$  or  $v_i \leq 0$  or  $j = i + 1$  then
8      $i \leftarrow i + 1$ ; continue
9   end
10  Choose  $\theta$  from nearest valid GPS points
11   $\Delta T \leftarrow t_j - t_i$ 
12  for each  $\Delta t$  in  $\{step, 2 \cdot step, \dots, < \Delta T\}$  do
13     $S \leftarrow v_i \Delta t$ ;
14     $(\hat{lat}, \hat{lon}) \leftarrow \text{dest\_point}(lat_i, lon_i, \theta, S)$ 
15    Append  $(\hat{lat}, \hat{lon}, t_i + \Delta t)$  to  $\mathcal{L}$ 
16  end
17  Append sample  $j$ ;
18   $i \leftarrow j$ ;
19 end
20 return  $\mathcal{L}$ 

```

This interpolation method produces a temporally dense and spatially smooth trajectory that aligns more faithfully with the vehicle's motion than the raw GPS log. Fig. 3 illustrates a representative segment: the original GPS trace contains sparsely distributed samples and repeated coordinates due to intermittent latency, whereas the reconstructed trace resolves these artifacts and follows the underlying road geometry with higher fidelity. This improvement

enhances the accuracy of travelled-distance estimation, ensures consistent synchronization between IMU and GPS streams prior to segmentation, and supports downstream tasks such as localized anomaly detection.

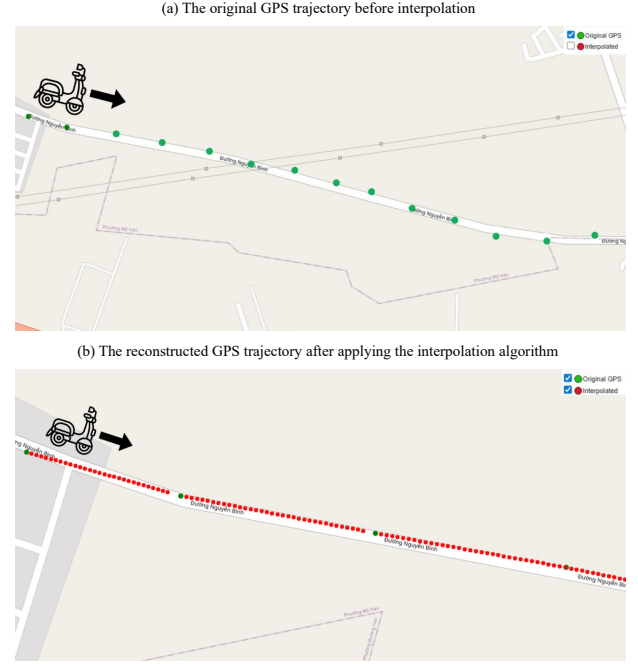


Fig. 3. Comparison of trajectory reconstruction results

F. Pothole detection using abnormality index

The interpolation mechanism in Section III-E restores spatial continuity during intervals of delayed or frozen GPS updates. This continuity is essential when analyzing short-duration disturbances whose locations must be assigned with high spatial precision. Pothole events fall within this category: they occur over very small road segments, and even minor GPS misalignment may shift their estimated positions. In this context, pothole detection provides a practical setting in which the benefits of interpolation become apparent.

A lightweight abnormality-index method is used to identify these events from triaxial accelerometer measurements. For each axis, a rolling mean filters out low-frequency trends, allowing local deviations to be highlighted. Let $a_x(n)$, $a_y(n)$, $a_z(n)$ denote the raw accelerations at sample n , and let $\bar{a}_x(n)$, $\bar{a}_y(n)$, $\bar{a}_z(n)$ be their corresponding rolling means. The abnormality index is then defined as:

$$AbI(n) = \sqrt{\begin{aligned} &[a_x(n) - \bar{a}_x(n)]^2 \\ &+ [a_y(n) - \bar{a}_y(n)]^2 \\ &+ [a_z(n) - \bar{a}_z(n)]^2 \end{aligned}} \quad (4)$$

A detection threshold is obtained from the global mean and standard deviation of the index:

$$Th = \mu_{AbI} + k \cdot \sigma_{AbI}, \quad (5)$$

where k is a scalar that controls the balance between sensitivity and robustness. Samples satisfying $AbI(n) > Th$ are grouped into contiguous intervals, each representing a candidate pothole event. For an interval $[s, e]$, the representative moment is identified as:

$$n_{\text{peak}} = \arg \max_{n \in [s, e]} AbI(n), \quad (6)$$

which corresponds to the time instant at which the abnormality index attains its maximum value within that interval, indicating the point of strongest vibration response.

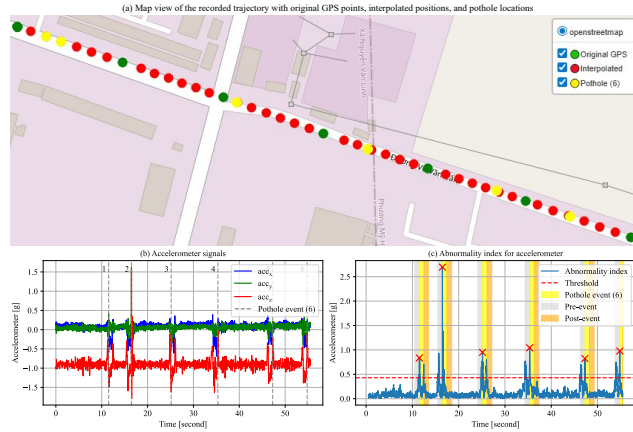


Fig. 4. Detection of pothole events using the abnormality index aligned with interpolated GPS positions

Fig. 4 illustrates the detection process, including raw accelerometer signals, the abnormality index, threshold crossings, and the localization of events along the interpolated trajectory. The integration of the interpolation mechanism with the abnormality-based detection algorithm ensures that highly localized disturbances are assigned spatial coordinates that are consistent with the vehicle's actual path. This interaction reinforces the methodological coherence of the pipeline: interpolation provides reliable positional grounding, and the detection process leverages this enhanced trajectory to support accurate spatial mapping of localized surface anomalies.

G. Evaluation on the private dataset

The private dataset provides a controlled evaluation setting for the MCU-oriented classification pipeline. Because the data were collected by a dedicated inspection fleet operating under uniform mechanical conditions, the classification task primarily reflects the intrinsic separability of the road-surface classes rather than cross-context variability.

A detailed characterization of the classification behaviour is shown in the per-class report in Table II. For dirt-road segments, XGBoost correctly identifies 3088 out of 3258 windows, corresponding to a recall of 94.78 %. Most misclassified cases are assigned to the cobblestone class (166 windows), with very limited confusion involving

asphalt (4 windows). Cobblestone represents the most challenging class; its vibration patterns exhibit irregular structures with partial similarity to those of dirt roads. As a result, XGBoost correctly recognizes 1950 out of 2157 cobblestone windows (recall 90.40 %), with misclassifications predominantly shifting toward dirt. Asphalt displays the most distinct signature among the three classes and is classified with high reliability: 3588 out of 3624 asphalt windows are correctly predicted, yielding a recall of 99.01 % and an F1-score of 98.57 %.

Overall, the results on the private dataset indicate that the MCU-constrained pipeline maintains strong discriminative capability despite stringent memory limitations. The comparable behaviour of XGBoost and LGBM suggests that tree-based models effectively capture the vibration characteristics relevant to road-surface discrimination while remaining feasible for deployment on embedded hardware. The per-class patterns align with intuitive difficulty levels, with cobblestone exhibiting the greatest ambiguity, asphalt being the easiest to distinguish, and dirt occupying an intermediate position.

TABLE I
PERFORMANCE AND COMPUTATIONAL COST OF CONSTRAINED CML MODELS ON THE PRIVATE DATASET

Model	Sensor	Configuration	HPO	Acc. (%)	Flash (kB)	Training time (s)	Prediction time (s)
XGBoost	Center	n_estimators ∈ [1, 50] max_depth ∈ [1, 20]	(41, 6)	95.43	975.17	0.5728	0.0054
RF			(8, 14)	92.82	888.38	0.4160	0.0061
LGBM			(50, 13)	95.02	893.15	1.6681	0.0131
ET			(6, 14)	91.92	936.08	0.0459	0.0054

Table I summarizes the accuracy, memory footprint, and computational cost of the four CML model families when constrained by the 1 MB flash budget. Among the evaluated methods, XGBoost attains the highest NCV accuracy at 95.43 ± 0.62 %, with a header footprint of 975.17 kB, remaining within the deployment limit. LGBM achieves a comparable accuracy of 95.02 %, whereas RF and ET yield 92.82 % and 91.92 %, respectively. Despite the restricted search space—limiting the number of trees to at most 50 and the maximum depth to 20—the reported accuracies fall within the upper range typically observed for classical models in the road-surface classification literature. Training times remain below 2 s for all models, and prediction times range between 5 ms and 10 ms per window on the evaluation CPU, satisfying the real-time requirements of the ESP32 given the 3 s stride of the sensing pipeline.

H. Evaluation of GPS interpolation and pothole detection

The spatial accuracy of the proposed framework is evaluated by comparing detected coordinates against a static ground truth. This reference was established by recording GPS coordinates while remaining stationary at each pothole location to obtain a stable position. During test runs at average speeds of 20 - 50 km/h, the localization error is measured as the geodesic distance between this static ground truth and two distinct points: (i) the raw GPS

TABLE II
CLASSIFICATION REPORT OF CML MODELS ON THE PRIVATE DATASET

Model	Class	Precision	Recall	F1-score
XGBoost	Dirt road	95.25	94.78	95.02
	Cobblestone road	91.08	90.40	90.74
	Asphalt road	98.14	99.01	98.57
RF	Dirt road	91.92	91.44	91.68
	Cobblestone road	85.39	86.74	86.06
	Asphalt road	98.14	97.68	97.91
LGBM	Dirt road	94.70	94.35	94.53
	Cobblestone road	90.28	89.57	89.92
	Asphalt road	98.08	98.87	98.47
ET	Dirt road	89.94	91.13	90.53
	Cobblestone road	84.88	83.03	83.95
	Asphalt road	97.82	97.93	97.88

coordinates and (ii) the nearest interpolated coordinates assigned at the moment of impact. This procedure provides a quantitative measure of how closely the reconstructed trajectory aligns with the physical pothole position compared to raw satellite logs.

TABLE III
COMPARISON OF POTHOLE LOCALIZATION ERRORS BETWEEN RAW GPS AND THE PROPOSED VELOCITY-AIDED INTERPOLATION ACROSS VARIOUS AVERAGE SPEEDS

Speed	20 km/h		30 km/h		50 km/h	
	Normal	Proposed	Normal	Proposed	Normal	Proposed
1	4.849	1.474	7.939	2.804	11.766	3.709
2	5.702	1.677	8.308	2.494	11.667	3.431
3	5.856	1.906	7.902	2.383	11.014	3.278
4	5.840	2.001	8.233	2.209	11.766	3.480
5	5.318	1.564	7.841	2.433	11.667	3.692
6	5.553	1.969	9.593	3.748	11.505	3.610
7	5.751	2.222	9.924	3.045	13.060	4.090
8	5.702	1.822	8.331	2.532	13.074	3.401
9	5.553	1.809	8.993	2.793	12.068	3.581
10	5.587	1.908	9.495	3.884	11.344	3.371

Normal: without interpolation; Proposed: interpolation method was applied

Table III summarizes the error metrics, including the mean error and root-mean-square error (RMSE). At 20 km/h, the raw GPS exhibits a mean error of 5.571 m, which is reduced to 1.835 m through interpolation. This improvement becomes more substantial as the vehicle speed increases and the raw GPS resolution gap widens. Specifically, the mean error decreases from 8.656 m to 2.833 m at 30 km/h, and from 11.893 m to 3.563 m at 50 km/h. Across all evaluated speeds, the proposed method significantly lowers the RMSE, notably decreasing from 11.911 m to 3.571 m at the 50 km/h threshold. These results indicate that the proposed interpolation method reduces the spatial gaps inherent in low-frequency updates, thereby improving the coordinate estimation as travel velocities increase.

IV. CONCLUSION

This research developed an IoT-based framework for RSC using a lightweight feature set and memory-constrained ML. The results show that optimized classical models achieve performance comparable to DL architectures while requiring significantly fewer computational

resources. This efficiency allows the system to operate in real-time on low-cost MCU. The framework also integrates GPS interpolation and pothole detection algorithms to improve trajectory reconstruction and hazard identification. Future work will focus on a real-time system to provide road-condition maps and automated alerts. This system will assist authorities in maintenance planning and provide road users with timely information to improve travel safety.

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